University of Virginia Library Interlibrary Loan

Electronic Delivery Cover Sheet

WARNING CONCERNING COPYRIGHT RESTRICTIONS

The copyright law of the United States (Title 17, United States Code) governs the making of photocopies or other reproductions of copyrighted materials.

Under certain conditions specified in the law, libraries and archives are authorized to furnish a photocopy or other reproduction. One of these specified conditions is that the photocopy or reproduction is not to be "used for any purpose other than private study, scholarship, or research". If a user makes a request for, or later uses, a photocopy or reproduction for purposes in excess of "fair use", that user may be liable for copyright infringement.

This institution reserves the right to refuse to accept a copying order if, in its judgment, fulfillment of the order would involve violation of copyright law.

Interlibrary Loan (ILL) is for individual research, scholarship, and personal study only. Under U.S. copyright law (Title 17, Sections 107 & 108), ILL materials (items not owned by UVA Library) cannot replace course materials or required readings*. Faculty and instructors seeking to provide course content not within the library's collection should work with your department Subject Liaison Librarian to explore appropriate alternatives. For more on Fair Use, see the Instructional Sharing page.*

Oxidation and mechanical behavior of intermetallic alloys in the Ti–Nb–Al ternary system

Guoliang Chen, Zuqing Sun and Xing Zhou
University of Science and Technology Beijing, Beijing 100083 (China)

Abstract

The isothermal oxidation behavior in a temperature range from 900 to 1100 °C and mechanical properties of intermetallic alloys with a wide range of compositions in the Ti–Nb–Al system were investigated. A composition range of the alloys having excellent oxidation resistance is suggested. The oxidation resistance maps summarizing the results of weight gain data and the constituents of oxide scale were established, as were the compressive strength maps as well. The lowest weight gain was determined to be 1–2 mg cm^{-2} and 2–3 mg cm^{-2} for oxidation at 1100 °C and 1200 °C for 100 h respectively. The compressive yield strength at 1100 °C and strain rates of 10^{-2} s^{-1} and 10^{-5} s^{-1} reached 450 MPa and 90 MPa respectively.

1. Introduction

The Ti–Nb–Al ternary system has recently received attention as a possible potential intermetallic structure system. It is not only because niobium addition improves the properties of Ti_3Al and TiAl compounds, but also because the ternary intermetallic alloys based on mixed phases of γ, Ti_3Al and γ_1 ($NbTiAl_3$) possess very attractive properties such as a high melting point and low density. Chen *et al.* [1, 2] proved the existence of the γ_1 phase and indicated that it was the result of selective distribution of niobium atoms in the $L1_0$ structure. The purpose of the paper is to understand the oxidation behavior and high temperature strength of alloys in the Ti–Nb–Al system in order to find the area with attractive oxidation and mechanical properties. The mechanism of oxidation is also discussed.

2. Experimental procedure

A wide range of compositions among the Ti_3Al, TiAl, $NbTiAl_3$, $NbAl_3$ and $TiAl_3$ intermetallics in the Ti–Nb–Al system was chosen for study. The specimens were prepared by vacuum induction melting (VIM) and casting to a preheated graphite mold. Homogenization was performed at temperatures as high as 1200–1380 °C. Static isothermal oxidation tests were conducted in air at temperatures from 900 to 1200 °C for 100 h. The X-ray diffraction (XRD) analysis of oxide scale was performed for all alloys. The oxide layers were analyzed by scanning electron microscopy (SEM) and electron microprobe analysis (EMPA). Compressive tests were performed on a Gleeble machine at various temperatures up to 1100 °C at strain rates of 10^{-2} and 10^{-5} s^{-1} in air.

3. Results

3.1. Oxidation behavior

For most alloys, the weight gain data in static air at temperatures from 900 to 1200 °C follow parabolic kinetics, and increase with increasing temperature. However, two alloys with high niobium contents exhibit linear kinetics and an abnormal decrease of weight gain at 1200 °C. Figure 1 shows an example of the effect of aluminum on the oxidation resistance.

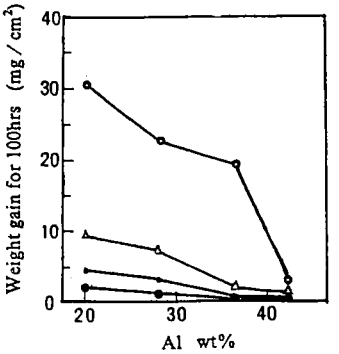


Fig. 1. Effect of aluminum content on the oxidation resistance of alloys (the niobium content is constant and equal to 28.10 wt.%): ○, 1200 °C; △, 1100 °C; •, 1000 °C; •, 900 °C.

0921-5093/92/$5.00

© 1992 – Elsevier Sequoia. All rights reserved

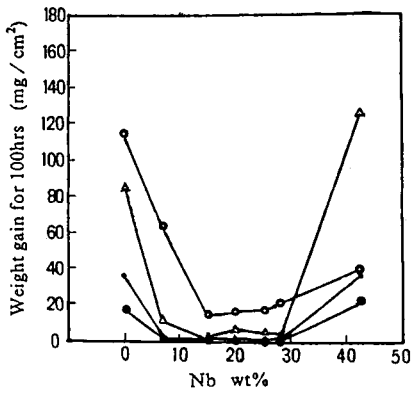


Fig. 2. Effect of niobium content on the oxidation resistance of alloys (the aluminum content is constant and equal to 36.50 wt.%): ○, 1200 °C; △, 1100 °C; •, 1000 °C; ●, 900 °C.

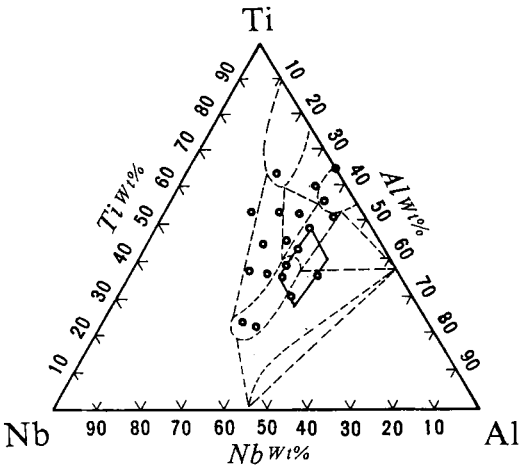
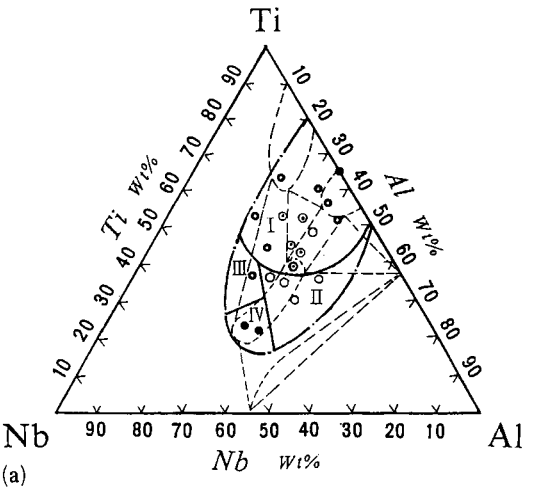


Fig. 3. The composition area having optimum oxidation resistance.



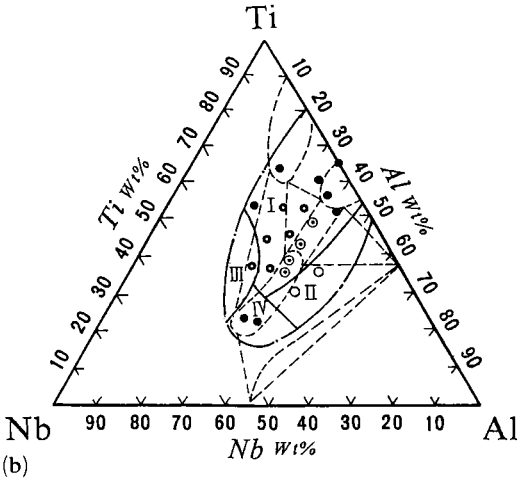


Fig. 4. Isothermal oxidation resistance maps for the area concerned in Ti-Nb-Al ternary system at (a) 1100 °C; •, 84.0-125.0 mg cm^{-2}; **O**, 7.0-15.0 mg cm^{-2}; ⊙, 4.0-7.0 mg cm^{-2}; ○, 1.7-3.0 mg cm^{-2}. (b) 1200 °C; •, 26.0-75.0 mg cm^{-2}; **O**, 20.0-25.0 mg cm^{-2}; ⊙, 12.0-20.0 mg cm^{-2}; ○, 2.0-4.0 mg cm^{-2}. (a) Oxide scale: I, $TiO_2 + \alpha\text{-}Al_2O_3$; II, $k\text{-}Al_2O_3 + TiO_2 + \alpha\text{-}Al_2O_3$; III, $TiO_2 + \alpha\text{-}Al_2O_3 + AlNbO_4$; IV, $Nb_2O_5 + TiO_2 + \alpha\text{-}Al_2O_3Al_2O_3 + \beta\text{-}Nb_2O_5$. (b) Oxide scale: I, $TiO_2 + \alpha\text{-}Al_2O_3$; II, $k\text{-}Al_2O_3 + \alpha\text{-}Al_2O_3 + TiO_2$; III, $TiO_2 + \alpha\text{-}Al_2O_3 + AlNbO_4$; IV, $Nb_2O_5 + TiO_2 + \alpha\text{-}Al_2O_3 + k\text{-}Al_2O_3$.

With increasing aluminum content from 28 to 37 wt.%, the weight gain of the alloys with various niobium contents from 15 to 28 wt.% decreases. Figure 2 shows the influence of niobium on the weight gain of the alloys with various aluminum content. Since the aluminum contents of the alloys in Fig. 2 are the same, the *X*-axis can actually be converted to Ti/Nb. It can be seen that very high Ti/Nb as well as very low Ti/Nb reduces the oxidation resistance significantly. Figure 3 shows the composition area with excellent oxidation resistance. The composition limitation can be described as: Al = 55-64 at.% and Ti/Nb (at.%) = 2-5. The weight gains of the alloys located in the area at 1100 °C and 1200 °C for 100 h are about 1-5 mg cm^{-2} and 3-20 mg cm^{-2} respectively.

Oxide constituents identified include TiO_2, $\gamma\text{-}Al_2O_3$, $k\text{-}Al_2O_3$, Nb_2O_5 and $AlNbO_4$ etc. Except for the typical TiO_2 oxide, the XRD pattern shows another pattern, which can be identified to be the same pattern as TiO_2 but has different lattice parameters. We believe it to be $(TiNb)O_2$. The oxidation resistance maps (Fig. 4(a) and (b)) summarize all information of the oxidation level, the oxide constitution and the effect of composition in the Ti-Nb-Al system. The oxidation resistance directly relates to the structure of the oxide scale and the relative amount of Al_2O_3. When Nb_2O_5 is present due to high niobium content, the oxidation resistance decreases significantly. Other oxides of niobium such as $AlNbO_4$ are not protective, but better than Nb_2O_5. The alloys in areas III and IV of the oxide scale in Fig. 4(a) and (b), therefore, are not satisfied.

The oxidation resistance of the alloys in area II of the oxide scale in the Fig. 4(a) and (b) is excellent since the relative amount of Al_2O_3 is predominant. The oxidation resistance of the alloys in the area I of the oxide scale in the Fig. 4(a) and (b) depends on the composition and the relative amount of Al_2O_3. A semi-quantitative analysis illustrates an obvious correlation between the weight gain and the relative amount of Al_2O_3 in terms of the relative ratio of the XRD intensity of total Al_2O_3 oxides to all the types of oxides in the oxide scale. Three groups can be distinguished. The first group includes the alloys with an aluminum content less than 45 at.%. The ratio of Al_2O_3 to total oxides is less than 0.2–0.25. The second group of alloys contains rather high aluminum contents (45–55 at.%). The ratio of Al_2O_3 increases to 0.25–0.6 depending on the ratio of Ti/Nb. If the ratio of Ti/Nb is too high or too low, the ratio of Al_2O_3 is less than 0.3. The oxidation resistance of these alloys, therefore, depends on both the aluminum content and the Ti/Nb ratio. The alloys with high aluminum contents, *e.g.* up to 64 at.%, belong to the third group of alloys. The Al_2O_3 becomes predominant and the oxidation resistance is excellent. The critical relative amount of Al_2O_3 for good oxidation resistance increases as the oxidation temperature increases.

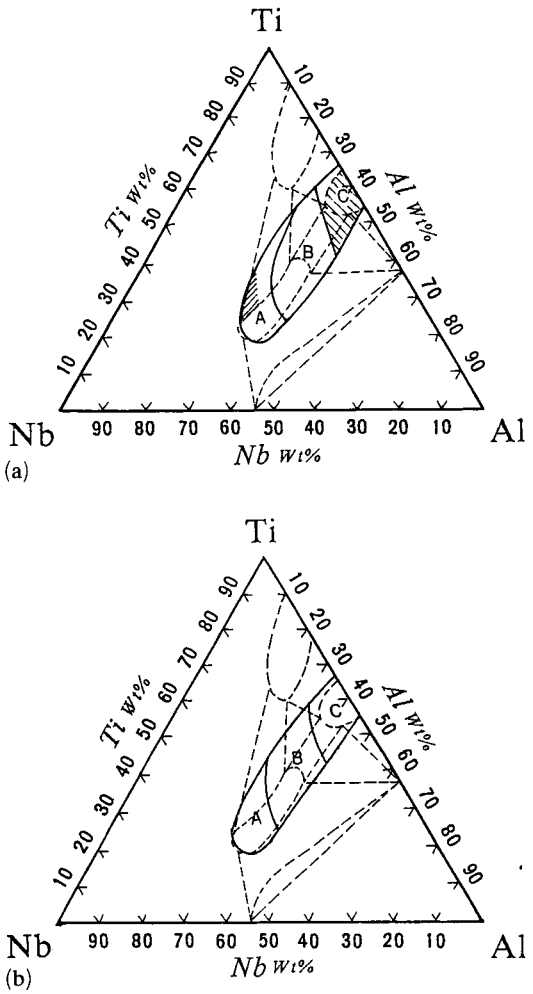


Fig. 5. High temperature compressive yield strength maps for the area concerned in Ti-Nb-Al ternary system at 1100 °C and strain rates of (a) 10^{-2} s^{-1} and (b) 10^{-5} s^{-1}. (a) A, 330–450 MPa; B, 200–330 MPa; C, 100–200 MPa. Fractography of tensile test at 1100 °C: ▨, trans-granular; ▨, intergranular; □, mixed transgranular and intergranular. (b) A, 60–100 MPa; B, 39–60 MPa; C, 28–39 MPa.

3.2. *Mechanical properties*

The results of high temperature strength measurements are summarized by the high temperature strength maps (Fig. 5(a) and (b)). It shows three areas possessing different strength levels. In general, the high temperature yield strength of the alloys based on $NbTiAl_3 + Ti_3Al$ or $NbTiAl_3 + (Nb,Ti)Al_3$ phases is always higher than (approximately twice) that of alloys based on the TiAl phase. The highest values of the high temperature strength at 1100 °C and strain rates of 10^{-2} s^{-1} and 10^{-5} s^{-1} reach 450 MPa and 90 MPa respectively. Fractography commonly reveals transgranular fracture. However, as the strength of the alloys increases, the fraction of the intergranular fracture increases for the 1100 °C tests at a strain rate of 10^{-2} s^{-1}.

4. Discussion

It was clearly shown that the structure of oxide scale is layered due to selective oxidation. Figure 6 shows a high magnification micrograph of a cross-section of an oxide scale. Table 1 shows micro-analysis data for various oxide layers and phases. The outer oxide layers include a dominant $TiO_2 + Al_2O_3$ layer (area 5) and a mixed layer of $(Ti,Nb)O_2 + Al_2O_3$ (area 4). Next is a dominant Al_2O_3 layer (area 3). A layer depleted in

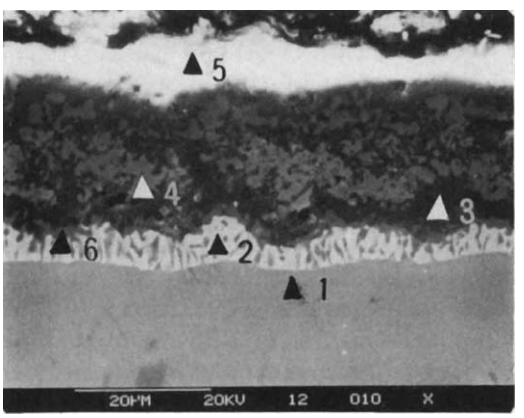


Fig. 6. Scanning electron micrograph of the morphology of the oxide scale for oxidation at 1100 °C for 100 h.

TABLE 1. Micro-analysis results for the oxide scale

Area[a]	Al (at.%)	Nb (at.%)	Ti (at.%)
Matrix	59.82	11.09	29.10
1	55.57	18.19	28.25
2	36.04	44.01	19.95
6	35.04	26.44	38.52
3	96.86	0.72	2.42
4	30.06	22.71	47.23
5	31.62	1.54	66.84

[a]See Fig. 6.

aluminum often appears next to the boundary between the matrix and the dominant Al_2O_3 layer. This layer includes a large white area enriched in niobium (area 2) and a small gray area enriched in titanium and niobium (area 6). We believe the corresponding phase to be $(Nb,Ti)_2Al$ (area 2). The outer dominant TiO_2 layer is always non-compact and non-protective. The integrity and thickness as well as the structure and defects of Al_2O_3 constituents within the Al_2O_3 layer control the oxidation resistance of the alloys. The intensity ratio of Al_2O_3 oxides and the distinction of α-Al_2O_3 and k-Al_2O_3 represent these factors. It can be seen that when the integrity of Al_2O_3 layer in Fig. 6 is good, the corresponding ratio of Al_2O_3 is about 0.5 and the weight gain for the 1100 °C test for 100 h is 4 mg cm^{-2}.

The oxide scale of the alloys with high niobium content (20–22 at.%) and with low aluminum content (36–44 at.%) has been analyzed in a similar way. The results indicate that the process is similar but Nb_2O_5 appears in every oxide layer for the alloys with high niobium, which makes the oxide porous and destroys the integrity of Al_2O_3 layer, and leads, we believe, to the poor protection of the oxide scale. For the alloys with low aluminum content, the $(Ti,Nb)O_2 + Al_2O_3$ layer is the main protective layer since the amount of Al_2O_3 is not enough to form a single Al_2O_3 layer.

According to the equation developed by Meier [3], the critical atom fractions of aluminum for forming a continuous Al_2O_3 layer for Ti–Al and Nb–Al system are larger than 59 at.% and 100 at.% respectively. Therefore, only for the alloys with high aluminum content and suitable Ti/Nb ratio (previous third group alloys) is a continuous Al_2O_3 layer able to form. The oxidation resistance is excellent. For most alloys, this is not the case. Rahmel [4] indicated by a thermodynamic calculation of oxide in the Ti–Al–O system that the selective oxidation of titanium could occur for the alloys with an aluminum content of less than 50 at.%. The selective oxidation of titanium leads to a sublayer abounding in aluminum. Since the oxygen atoms easily diffuse inwards for the TiO_2 oxide with a large amount of anion vacancies, the Al_2O_3 oxide forms under the outer oxide layer, and it causes the formation of $(Nb,Ti)_2Al$ at one side of the Al_2O_3 layer and the formation of $(Ti,Nb)O_2$ at the other side. We believe that Nb^{5+} substituting for Ti^{4+} should reduce the anion vacancies and improve the protection of the TiO_2 oxide. The increase of oxidation temperature promotes the selective oxidation of titanium and reduces the amount of Al_2O_3. Therefore, the critical aluminum content for best oxidation resistance must be raised. It can be proved by comparing Fig. 4(a) with Fig. 4(b).

The high temperature strength maps (Fig. 5(a) and (b)) indicate that niobium addition increases the high temperature strength and the trend of intergranular fracture. The microstructure of the alloys affects the high temperature strength significantly. The strength of the alloys based on the TiAl phase is always lower than that of alloys based on the $NbTiAl_3$ phase. The multiphase structure often offers better mechanical properties. An abnormal temperature dependence of compressive flow stress occurs for all alloys. It should be pointed out that the composition region with the best oxidation resistance is not consistent with the region with the highest high temperature strength. More work must be done to further improve the properties of the alloys.

5. Conclusions

(1) The composition area with excellent oxidation resistance in the Ti–Nb–Al system can be described as: Al = 55–64 at.% and Ti/Nb (at.%) = 2–5. The lowest weight gain data for two alloys with high aluminum content are determined to be 1–2 mg cm^{-2} and 2–3 mg cm^{-2} for oxidation at 1100 °C and 1200 °C for 100 h respectively.

(2) The integrity and thickness as well as the structure and defects of the Al_2O_3 constituent within the Al_2O_3 layer control the oxidation resistance of the alloys. The $(Ti,Nb)O_2 + Al_2O_3$ layer within the oxide scale is also protective while the TiO_2 is not considered to be protective. The occurrence of Nb_2O_5 always leads to a decrease in the oxidation resistance. The oxidation resistance maps summarize all the information on the oxidation level, the scale constituents and the effect of composition in the Ti–Nb–Al phase diagram. The oxidation mechanism closely relates to the selective oxidation behavior of titanium, niobium and aluminum, which are variable for the alloys with different compositions.

(3) The high temperature strength maps show the level of the yield strength at 1100 °C and strain rates of 10^{-2} and 10^{-5} s^{-1} for the alloys studied is very high. In particular, it should be noted that the density of the alloys is about the same as Ti_3Al alloys. However,

the composition area for the best oxidation resistance does not necessarily agree with the composition area for the best mechanical properties.

Acknowledgments

Research was sponsored by NAMCC under contract 863-715-16-02-09. Part of the work was supported by NNSFC.

References

1 G. Chen, Z. Sun, X. Xie, Y. Ren, Y. Xu, K. Yao, X. Zhou, L. Sha, S. Fu and W. Yang, *Proc. C-MRS International '90*, Vol. 2, Elsevier, Amsterdam, 1991, p. 803.
2 G. Chen, K. Yao, Z. Sun, Y. Xu and Y. Ren, submitted for publication.
3 G. H. Meier, in T. Grobstein and J. Doychak (eds.), *Oxidation of High Temperature Intermetallics*, TMS, Warrendale, PA, 1989, p. 1.
4 A. Rahmel and P. J. Spencer, *Oxid. Met., 35* (1991) 53.